

DEYUNG SUN

Mechanical Engineer | Equipment Development & Engineering Testing

U.S. Citizen (Taiwan-based) | Fluent in English and Mandarin

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SUMMARY

Mechanical Engineering M.S. student/graduate with experience in mechanical design, engineering simulation, system testing, and equipment troubleshooting. Skilled in SolidWorks, AutoCAD, Python, and MATLAB, with experience in mechanical prototyping, functional testing, and field troubleshooting of electromechanical systems. Experienced in developing simulation models, designing repeatable engineering tests, analyzing performance data, and documenting technical results. Seeking a Mechanical Engineer role focused on equipment development, engineering analysis, testing, and continuous improvement.

EDUCATION

California State Polytechnic University, Pomona Master of Science in Mechanical Engineering | Jun 2024 - Jun 2026 | GPA: 3.75/4.0

National Cheng Kung University, Taiwan Bachelor of Science in Mechanical Engineering | Sep 2019 - Jun 2023

EXPERIENCE

System Engineer (Equipment Technical Support) — Proaction Technology Inc. Taiwan /
Jun 2026 - Present

- Support field testing, technical evaluation, and troubleshooting of industrial equipment and associated control systems
- Analyze equipment operating parameters, technical documentation, and field test results to identify potential causes of system abnormalities
- Collect and compare operating and test data to develop troubleshooting hypotheses and plan verification tests
- Review OEM manuals, equipment specifications, and control documentation to support systematic problem diagnosis
- Communicate technical issues, field observations, and test results in English with U.S.-based OEM technical personnel

Research Assistant — Autonomous Vehicle Laboratory *California State Polytechnic University, Pomona* / Jun 2025 – Jun 2026

- Developed Gazebo simulation models based on the lab's physical mobile robot platform, replicating its kinematic configuration, dimensions, and sensor placement
- Configured simulated LiDAR, IMU, and odometry sensors to support navigation and control development in ROS 2
- Designed and executed repeatable simulation test scenarios to evaluate navigation and control performance under varied conditions
- Analyzed simulation outputs using Python and MATLAB to support parameter tuning, performance evaluation, and design iteration
- Documented simulation configurations, test procedures, observed failure modes, and corrective actions to support systematic engineering evaluation

Engineer — Proaction Technology Inc. *Taiwan* / Jul 2023 – Dec 2023

- Performed field deployment, maintenance, and troubleshooting of electromechanical monitoring systems
 - Diagnosed failures involving mechanical components, sensors, and wiring under field operating conditions
 - Executed functional testing, safety checks, and system commissioning
 - Restored system functionality through structured troubleshooting and verification procedures
 - Recorded equipment issues, corrective actions, and test outcomes to support subsequent maintenance and reliability improvement
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ENGINEERING PROJECTS

Campus Autonomous Driving Robot — ROS 2 / Gazebo

- Developed URDF/SDF robot models and simulation environments for autonomous navigation system evaluation
- Implemented and tested navigation, perception, and control workflows using ROS 2 and Gazebo
- Designed structured test scenarios and analyzed results to tune system parameters and improve performance

Logistics Warehouse Robot — Mechanical Design / LEGO EV3

- Designed and built a mobile robot incorporating drivetrain and gripping mechanisms
- Integrated mechanical components, actuation, and control functions and refined the system through iterative testing

- Implemented PID control and achieved Top 3 (Honorable Mention) among 27 engineering competition teams
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TECHNICAL SKILLS

Mechanical Design & Manufacturing: SolidWorks, AutoCAD, mechanical design, rapid prototyping, FDM 3D printing, DFM/DFA fundamentals, manual milling and lathe machining, exposure to 5-axis CNC

Engineering Analysis & Testing: Engineering testing, functional testing, system validation, failure analysis, troubleshooting, data analysis, test documentation

Programming & Simulation: Python, C++, MATLAB, ROS 2 (Jazzy), Gazebo, Linux, Git, URDF/SDF

Equipment & Systems: Electromechanical troubleshooting, field testing, commissioning, sensor and wiring diagnostics, technical documentation

Languages: Mandarin Chinese (Native), English (Fluent)